1A. $\mathbb{R}^n$ and $\mathbb{C}^n$

- $\mathbb{C} = \{ \underbrace{a+bi}_{\text{ordered pair}} : a, b \in \mathbb{R} \}$

- (add) $(a+bi) + (c+di) = (a+c) + (b+d)i$

- (mul) $(a+bi)(c+di) = (ac-bd) + (ad+bc)i$

i.e $i^2 = -1$

$\alpha, \beta, \lambda \in \mathbb{C}$

commutative	associative	identities	inverses
$\alpha+\beta = \beta+\alpha$	$(\alpha+\beta)+\lambda = \alpha+(\beta+\lambda)$	$\lambda + 0 = \lambda$	$\alpha + \underbrace{(-\alpha)}_{\text{unique}} = 0$
$\alpha\beta = \beta\alpha$	$(\alpha\beta)\lambda = \alpha(\beta\lambda)$	$\lambda \cdot 1 = \lambda$	$\alpha \cdot \underbrace{(1/\alpha)}_{\text{unique}} = 1$

distributive $\lambda(\alpha+\beta) = \lambda\alpha + \lambda\beta$

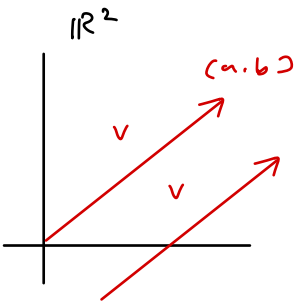
- $\mathbb{F}$ stands either $\mathbb{R}$ or $\mathbb{C}$
- elements of $\mathbb{F}$ are called scalar

- list of length n is ordered collection of n elements
- two lists are equal iff ① same length
  ② same elements in the same order.

- $\mathbb{F}^n = \{ \underbrace{(x_1, \ldots, x_n)}_{\text{list of length } n} : x_k \in \mathbb{F}^n \text{ for } k = 1, \ldots, n \}$

- add : $(x_1, \ldots, x_n) + (y_1, \ldots, y_n) = (x_1 + y_1, \ldots, x_n + y_n)$
- scalar mul : $\lambda(x_1, \ldots, x_n) = (\lambda x_1, \ldots, \lambda x_n)$

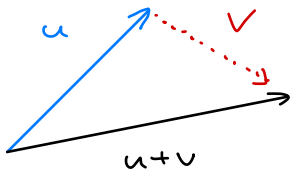
- (add identity) $0 = (0, \ldots, 0)$
- (add inverse) $-x = (-x_1, \ldots, -x_n)$



elements of $\mathbb{R}^2$

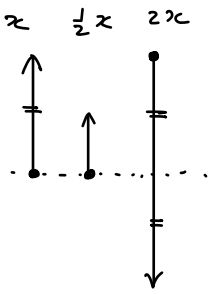
thought as ① points

② vectors



sum of

two vectors



scalar

mul.

1B. Def of Vector Space

define two functions on V

① add : $u+v \in V$ if $u, v \in V$

② scalar mul : $au \in V$ if $a \in \mathbb{F}$, $u \in V$

V is vector space over $\mathbb{F}$ iff ... $u, v, w \in V$, $a, b \in \mathbb{F}$

commutative

$u+v = v+u$

associative

$(u+v)+w = u+(v+w)$

$(ab)v = a(bv)$

identity

$v + \underbrace{0}_{\text{unique}} = v$

$v \cdot 1 = v$

add inverse

$v + \underbrace{(-v)}_{\text{unique}} = 0$

distributive

$a(u+v) = au + av$

$(a+b)v = av + bv$

- element of vector space are called vectors (or points)

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① real vector space : vector space over $\mathbb{R}$

② complex " " : " " " $\mathbb{C}$

$\mathbb{F}^S$ denotes set of functions $S \to \mathbb{F}$

(add) $(f+g)(x) = f(x) + g(x)$

(scalar mul) $(\lambda f)(x) = \lambda f(x)$

then ① $\mathbb{F}^S$ is vector space if $S \neq \emptyset$

② (add identity) $0(x) = 0$

③ (add inverse) $(-f)(x) = -f(x)$

---

[0]. $\mathbb{F}^n$, $\mathbb{F}^\infty$, $\mathbb{R}^{[0,1]}$ are vector spaces

$\mathbb{F}^n = \mathbb{F}^{\{1,\dots,n\}}$, $\mathbb{F}^\infty = \mathbb{F}^{\mathbb{N}}$

- unique add identity $\because$ $\mathbb{0}' = \mathbb{0}' + \mathbb{0}$

$$= \mathbb{0} + \mathbb{0}' = \mathbb{0}$$

- unique add inverse $\because$ $w = w + \mathbb{0}$

$$= w + (v + w')$$

$$= (w + v) + w'$$

$$= \mathbb{0} + w' = w'$$

- $\mathbb{0} v = \mathbb{0}$ $\because$ $\mathbb{0} v = (\mathbb{0} + \mathbb{0}) v = \mathbb{0} v + \mathbb{0} v$

- $(-1) v = -v$ $\because$ $v + (-1) v = 1 v + (-1) v$

$$= (1 + (-1)) v$$

$$= \mathbb{0} v = \mathbb{0}$$

- $a \mathbb{0} = \mathbb{0}$ $\because$ $a \mathbb{0} = a (\mathbb{0} + \mathbb{0}) = a \mathbb{0} + a \mathbb{0}$

1C. subspace

- $U \subseteq V$ is called subspace of V

$\Longleftrightarrow$ U is vector space with same
① add identity
② add
③ scalar mul on V

$\Longleftrightarrow$ U satisfies

(i) add identity $0 \in U$

(ii) closed under add $u + w \in U$ $(u, w \in U)$

(iii) closed under scalar mul $au \in U$ $(a \in \mathbb{F},\ u \in U)$

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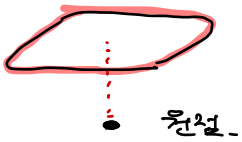
① $\{(\cancel{b}, 0, 0) + (0, x, y)\} \subseteq \mathbb{R}^3$

is subspace iff $b = 0$



2D plane.

② conti functions in $\mathbb{R}^{[0,1]}$ is subspace

③ diffable functions in $\mathbb{R}^{\mathbb{R}}$ is subspace

④ $\{$diffable $f : f'(2) = \cancel{b}\} \subseteq \mathbb{R}^{(0,3)}$

is subspace iff $b = 0$

⑤ $\{$seq with limit zero$\} \subseteq C^{\infty}$ is subspace

Sums of Subspaces

let $V_1, \dots, V_m$ are subspaces of U

then $V_1 + \cdots + V_m = \{ v_1 + \cdots + v_m : v_1 \in V_1, \dots, v_m \in V_m \}$

is the smallest subspace containing $V_1, \dots, V_m$

---

because $V_k \subseteq \overbrace{V_1 + \cdots + V_m}^{\text{subspace}} \subseteq$ subspace containing $V_1, \dots, V_m$

$0 + \cdots + v_k + \cdots + 0$

closed under add.

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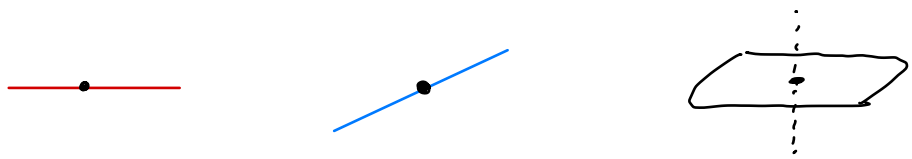
e.g $\{(x, 0, 0)\} + \{(0, y, 0)\} = \{(x, y, 0)\}$



1차원 1차원 2차원

e.g $\{(x, x, y, y)\} + \{(x, x, x, y)\} = \{(x, x, y, z)\}$

---

direct sum $V_1 \oplus \cdots \oplus V_m$ means

unique expression $v = v_1 + \cdots + v_m$ for $\forall v \in V_1 + \cdots + V_m$

$\Longleftrightarrow$ unique expression $0 = 0 + \cdots + 0$ for 0

$\because$ double expression $v = v_1 + \cdots + v_m$

$= w_1 + \cdots + w_m$

then $0 = (v_1 - w_1) + \cdots + (v_m - w_m)$

- $\{(x, y, z)\} = \{(x, y, 0)\} \oplus \{(0, 0, z)\}$
- $\mathbb{F}^n = \{(x, 0, 0, \ldots, 0)\} = V_1$

$\oplus \{(0, x, 0, \ldots, 0)\} = V_2$

$\cdots$

$\oplus \{(0, 0, 0, \ldots, x)\} = V_n$

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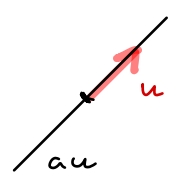
direct sum $U \oplus W$ iff $U \cap W = \{0\}$



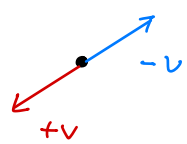
each vector implies a line

i.e $u \in U$ then $au \in U$

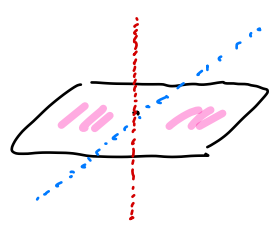
$U + W$ $\Longleftrightarrow$ $0 = v + (-v)$ $\Longleftrightarrow$ $\{0, \underbrace{-v, v}_{\text{closed under mul}}\} \subseteq U \cap W$

not direct sum

not unique



- $\mathbb{F}^3 = \{(x, y, 0)\}$

$+ \{(0, 0, z)\} + \{(0, y, y)\}$

is not direct sum